

1. Smart Multilingual Conferencing

Abstract:

Current video conferencing platforms like Zoom and Google Meet offer basic multilingual support using text subtitles and limited translations. However, these features fall short in delivering real-time, natural communication, especially for non-native speakers or users with accessibility needs. Moreover, they do not support real-time voice dubbing or code-switching, which are essential in multilingual environments. This project introduces an AI-powered conferencing web application that performs live voice translation by capturing audio using WebRTC, transcribing speech via Whisper, translating it through M2M-100 or NLLB, and generating spoken output using Text-to-Speech (TTS) models such as Bark or Coqui. The system supports low-latency streaming, multilingual transcription, and natural-sounding voice synthesis. Key enhancements include the ability to integrate emotion-aware dubbing, and support modular pipelines for efficient processing. By enabling seamless voice-based translation across languages, the app offers a more inclusive and human-like experience for international collaboration, virtual education, and global business. It surpass traditional subtitle-based systems, setting a new benchmark for real-time multilingual communication.

Keywords: Multilingual communication, real-time dubbing, Whisper, WebRTC, TTS, Bark, Coqui, code-switching, AI translation, voice synthesis.

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